

PIA 404 Course Outline

PIA 404 – Economic Cost-Benefit Analysis

What the resources are worth, and to whom.

School	04 – Public Investment
Hours	40 (indicative band 36-45)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	PIA 401 for the conventions and the governed parameter set, PIA 402 for incremental demand, and PIA 403 for discounting and the present value operations. PIA 406 may be taken before or after; this course uses conversion factors and PIA 406 derives them.
Audience	Appraisal economists, development committee secretariats, sector planning units, and reviewers who have to decide whether an economic case is defensible
Pass mark	60 %

What this course is

The economic appraisal asks whether the country is better off. It works in economic prices rather than market ones, removes transfers rather than counting them, uses the social discount rate rather than the financial one, and takes the perspective of the economy rather than of the entity holding the cash.

It produces five statistics and they do not agree with one another. In the sample register the highest ENPV, the highest benefit-cost ratio and the highest net benefit investment ratio are three different projects, and each is the right answer to a different question. This course is largely about knowing which question you are being asked.

It also produces a transformation table, and that table is the most honest artefact in the whole system. For the reference project it converts 11 cost categories, of which 8 have a governed conversion factor, 2 are provisions converted at the weighted average factor of the base cost they cover, and 3 have no governed factor at all. Those 3 are carried at financial value, counted in the unconverted share of 6.4%, and each carries the engine's own note: NO GOVERNED FACTOR – this is not an assumption that the market price equals the economic value.

The course's central discipline is that the choice of statistic is part of the recommendation. Reporting a benefit-cost ratio without saying that the capital constraint would have ranked differently is not a presentational choice; it is an omission.

What a graduate can do

Convert a financial cost profile into an economic one, category by category, and read the resulting transformation table including its unconverted share; explain why a provision converts at the weighted average factor of the base cost it covers rather than at a parameter somebody set; compute ENPV, EIRR, BCR, NBIR and FYRR and say which question each answers; choose the right ranking statistic under a hard capital constraint and demonstrate that it differs from the BCR; use the first-year rate of return to test whether a project is being built too early; identify what must be excluded from economic cost and why; read a sector benefit model and state what it refuses to derive; and write an economic case that says what it does not establish.

What this course does not claim

This course does not derive conversion factors. It applies the governed set and reads what the engine did with it; PIA 406 derives the factors, the shadow wage, the shadow exchange rate and the values of time and life.

It does not price risk. Every figure here is a point estimate at the governed social discount rate. PIA 407 tests it one variable at a time and finds the switching values; PIA 408 puts a distribution round it.

It does not appraise what cannot be priced. Where the outcome has no defensible money value the instrument is cost-effectiveness, which is PIA 405. One register project has a benefit-cost ratio of exactly 0.00 for that reason, and this course uses it as a boundary case rather than as a failure.

It does not supply sector parameters. The vehicle operating costs, the yield responses, the returns to schooling and the burden estimates are all governed values or sector-model outputs, and this course quotes them with their source rather than teaching how they were estimated.

Learning outcomes

1. State the difference between financial and economic appraisal, and what must be excluded from economic cost.
2. Convert a financial cost profile into an economic one, category by category, from the governed set.
3. Read a transformation table, including its unconverted share, and say what that share permits.
4. Explain why a provision converts at the weighted average factor of the base cost it covers.
5. Compute an economic net present value and an economic internal rate of return.
6. Compute a benefit-cost ratio and say what it is and is not sensitive to.
7. Compute a net benefit investment ratio and use it to rank under a hard capital constraint.
8. Demonstrate that the BCR and the NBIR rank the same cohort differently, and explain why.
9. Compute a first-year rate of return and use it to test whether a project is being built too early.
10. Choose the right statistic for a stated decision and justify the choice.
11. Read a sector benefit model and state what it computes and what it refuses to derive.
12. Apply the rule of half and the with-and-without principle to a benefit stream.
13. Identify a benefit that should not be counted, and say why.
14. Write an economic case that states what it does not establish.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	The economic question	1	1.25	–	1	Quiz M01
M02	From financial to economic cost	2	3.50	Lab 01	2-3	Quiz M02
M03	The transformation table	1	1.50	–	4	Quiz M03
M04	Provisions and the blended factor	1	1.50	–	5	Quiz M04
M05	ENPV and EIRR	2	3.50	Lab 02	6-7	Quiz M05
M06	The benefit-cost ratio	1	1.50	–	8	Quiz M06
M07	The net benefit investment ratio	1	3.50	Lab 03	9	Quiz M07
M08	When the statistics disagree	1	1.50	–	10	Quiz M08
M09	The first-year rate of return	1	1.50	–	11	Quiz M09
M10	Choosing the statistic	1	1.50	–	12	Quiz M10
M11	Sector benefit models	2	3.50	Lab 04	13-14	Quiz M11
M12	Transport, energy and water	1	1.50	–	15	Quiz M12
M13	What is not a benefit	1	3.50	Lab 06	16	Quiz M13
M14	Writing the economic case	1	3.50	Lab 05	17	Quiz M14

Laboratories

Lab	Title	Hours	Deliverable
1	Convert a cost profile, and reconcile the unconverted share	2.5	A transformation table with the share reconciled
2	Compute the five statistics, and check them against the engine	2.5	Five statistics, reconciled, and a reclassification demonstration
3	Rank the cohort three ways, and find where they disagree	2.5	Three rankings, the spreads, and a committee paragraph
4	Read a sector benefit model, and state what it refuses	2.5	A sector benefit note
5	Write the economic case, ending with what it does not establish	2.5	A complete economic case
6	Audit a benefit stream by category and valuation basis	2.5	The benefit audit

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The transformation table	20	Marked on the blended factor and the share's denominator
A2	The five statistics	25	Marked on the demonstration that the BCR moves and the ENPV does not
A3	Three rankings and a decision	30	Marked on stating the decision before choosing the statistic
A4	The economic case	25	Marked on the limits section and its named owners

Final project

A complete economic case for one sample project: the financial cost profile converted category by category with the unconverted share stated; the benefit streams with their valuation bases; ENPV, EIRR with its status, BCR, NBIR and FYRR; a statement of which statistic should drive the recommendation and why, demonstrated by showing that at least two of them rank the cohort differently; the exclusions applied and their justification; and a closing section on what the case does not establish.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.